Project 1

Contents

1.Understanding the database context
2.Creation of a pluggable database
3.User creation, privileges, roles and user profiles
4.Changing database parameters
5.Resources

1.Understanding the database context

First of all, we will take a look at some parameters on the database. To do this we will use Sql*Plus. We'll authenticate as sys user.

```
Introduzca el nombre de usuario: sys as sysdba
Introduzca la contrase±a:

Conectado a:
Oracle Database 23ai Free Release 23.0.0.0.0 - Develop, Learn, and Run for Free
Version 23.6.0.24.10

SQL> show user;
USER es "SYS"
SQL> show con_name;

CON_NAME
------------------------------
CDB$ROOT
SQL> show pdbs;

    CON_ID CON_NAME                       OPEN MODE  RESTRICTED
---------- ------------------------------ ---------- ----------
         2 PDB$SEED                       READ ONLY  NO
         3 FREEPDB1                       READ WRITE NO
         4 FREEPDB10                      READ WRITE NO
         5 PDBUNDO                        MOUNTED
         6 PDBX1                          MOUNTED
         7 PDBX2                          MOUNTED
         8 PDB_NEBULA                     READ WRITE NO
```

In the last picture we are checking in which database we are, with which user and finally we are showing all pdbs in our container database. To visualize the following information we will use the SqlDeveloper to have better readability.

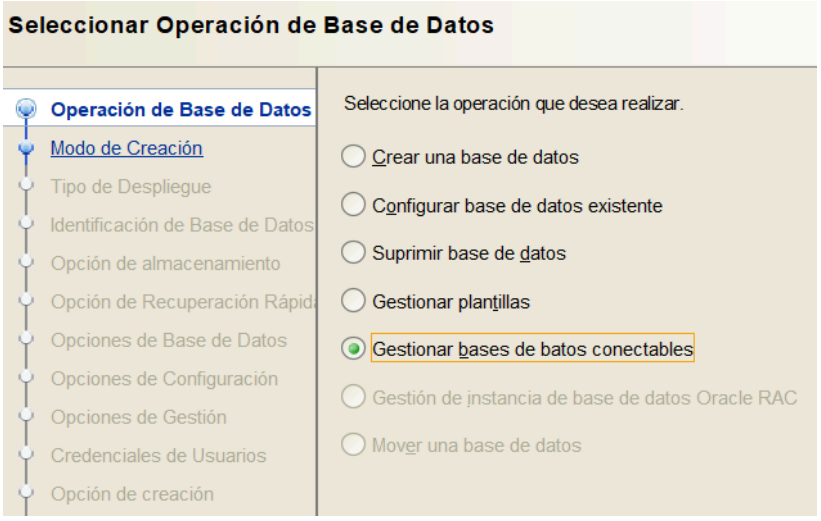
In the following image we are using dynamic views to see crucial information about our database such as the instance name, the container database, our pluggable databases and the database version.

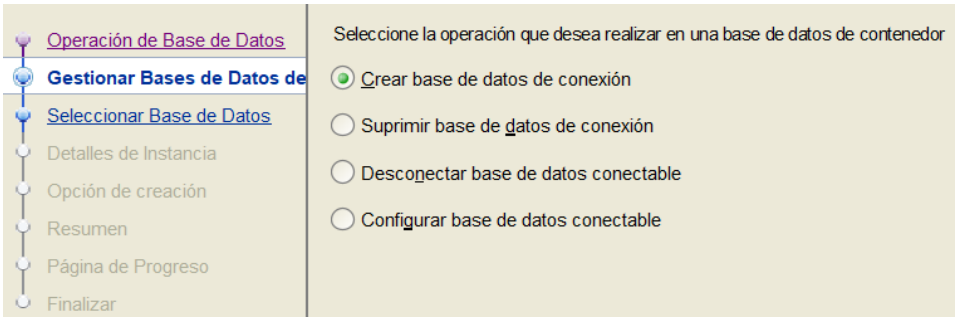
```sql
--Dynamic views

--Figure out the instance name
SELECT instance_name, host_name, status, edition
FROM v$instance;

--Visualize info about the container database
SELECT name, cdb, con_id, OPEN_MODE
FROM v$database;

--Check the database version
SELECT banner
FROM v$version;

--Check info about the cdb and pdbs
SELECT con_id, name, open_mode
FROM v$containers;

--Check info about the pdbs
SELECT name, open_mode, con_id
FROM v$pdbs;
```

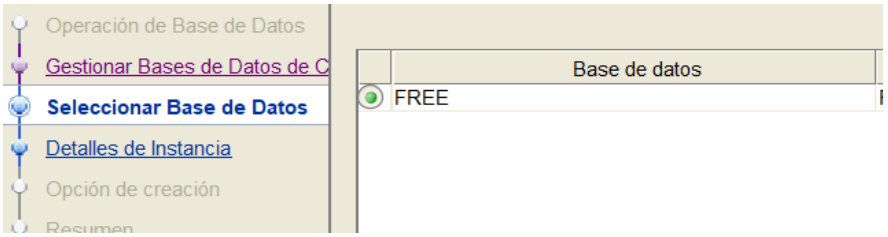
2.Creation of a pluggable database

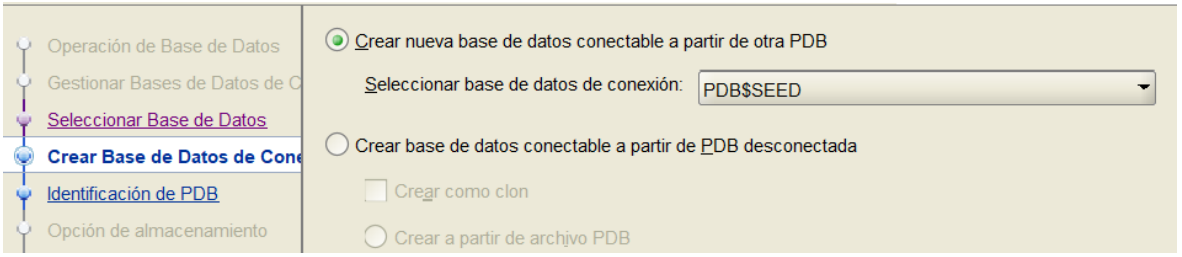
We will create new pbds by using different methods. First, we will use the Database Configuration Assistant also known as dbca to create a new pdb. This is a really easy way because it is a graphical tool and we only have to follow the steps and fill in any information.

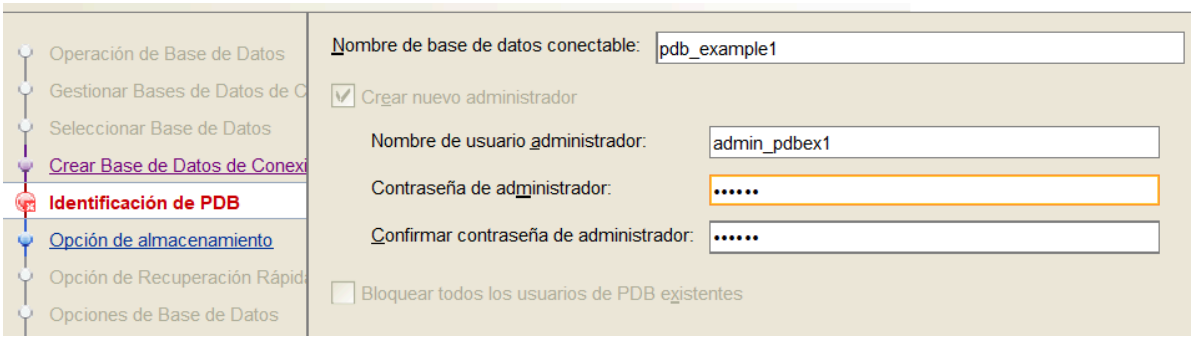
To startup the dbca we get into the terminal. We will go through the following screens:

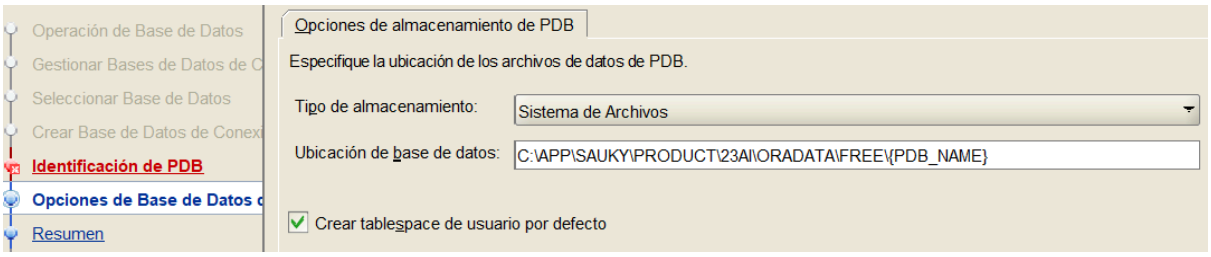












After we create this new pdb we will be able to see the physical archives or datafiles created by default in the following route: C:\app\sauky\product\23ai\oradata\FREE\pdb_example1.

SYSAUX01.DBF	03/03/2026 18:48	Hoja de cálculo d...	419.848 KB
SYSTEM01.DBF	03/03/2026 18:27	Hoja de cálculo d...	286.728 KB
TEMP01.DBF	03/03/2026 18:27	Hoja de cálculo d...	20.488 KB
UNDOTBS01.DBF	03/03/2026 18:27	Hoja de cálculo d...	102.408 KB
USERS01.DBF	03/03/2026 18:27	Hoja de cálculo d...	7.176 KB

Once we have the new pluggable database created let's check the listener and the services that were created for the pdb.

```
C:\Users\sauky>lsnrctl status

LSNRCTL for 64-bit Windows: Version 23.0.0.0.0 - Production on 04-MAR-2026 16:01:10

Copyright (c) 1991, 2024, Oracle.  All rights reserved.

Conectßndose a (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=miguelse)(PORT=1521)))
ESTADO del LISTENER
------------------------
Alias                     LISTENER
Versi¾n                   TNSLSNR for 64-bit Windows: Version 23.0.0.0.0 - Production
Fecha de Inicio       02-MAR-2026 11:41:08
Tiempo Actividad   2 dÝas 4 hr. 20 min. 1 seg.
Nivel de Rastreo        off
Seguridad               ON: Local OS Authentication
SNMP                      OFF
Parßmetros del Listener   C:\app\sauky\product\23ai\dbhomeFree\network\admin\listener.ora
Log del Listener          C:\app\sauky\product\23ai\dbhomeFree\log\diag\tnslsnr\miguelse\listener\alert\log.xml
```

Besides, we can see the service:

```
El servicio "pdb_example1" tiene 1 instancia(s).
  La instancia "free", con estado READY, tiene 5 manejador(es) para este servicio...
```

Let's move on to check out the method connections. If we try to connect by using the TNSnames.ora we notice the alias pdb_example1 wasn't found in the archive. There are many methods to create it, we can use for instance the Network Configuration Assistant, however in this case we will modify the tnsnames.ora manually.

```
C:\Users\sauky>sqlplus admin_pdbex1/oracle@pdb_example1

SQL*Plus: Release 23.0.0.0.0 - Production on MiÚ Mar 4 16:07:44 2026
Version 23.6.0.24.10

Copyright (c) 1982, 2024, Oracle.  All rights reserved.

ERROR:
ORA-12154: No se puede conectar a la base de datos. No se ha encontrado el
alias pdb_example1 en
C:\app\sauky\product\23ai\dbhomeFree\network\admin\tnsnames.ora.
Ayuda: https://docs.oracle.com/error-help/db/ora-12154/
```

```
PDB_EXAMPLE1 =
  (DESCRIPTION =
    (ADDRESS_LIST =
      (ADDRESS = (PROTOCOL = TCP)(HOST = miguelse)(PORT = 1521))
    )
    (CONNECT_DATA =
      (SERVICE_NAME = pdb_example1)
    )
  )
```

Finally, as we can see in the following image, we are capable of connecting to the database using the TNSnames.ora.

```
C:\Users\sauky>sqlplus admin_pdbex1/oracle@pdb_example1

SQL*Plus: Release 23.0.0.0.0 - Production on MiÚ Mar 4 16:18:43 2026
Version 23.6.0.24.10

Copyright (c) 1982, 2024, Oracle.  All rights reserved.

Hora de ┌ltima Conexi¾n Correcta: MiÚ Mar 04 2026 15:40:34 +01:00

Conectado a:
Oracle Database 23ai Free Release 23.0.0.0.0 - Develop, Learn, and Run for Free
```

Now, we are going to create a pdb using sql. Firstly, we will create a new directory to place the datafiles: C:\app\sauky\product\23ai\oradata\FREE\pdb_example2.

```
--Creating a new pdb
CREATE PLUGGABLE DATABASE pdb_example2 --naming the new pdb
  ADMIN USER admin_pdbex2 IDENTIFIED BY oracle --admin user for this pdb
  ROLES = (dba) -- giving this role to the admin user
  DEFAULT TABLESPACE users --give a default table space
    DATAFILE 'C:\app\sauky\product\23ai\oradata\FREE\pdb_example2\users01.dbf' SIZE 250M AUTOEXTEND ON
  FILE_NAME_CONVERT = ('C:\app\sauky\product\23ai\oradata\FREE\pdbseed\',
                       'C:\app\sauky\product\23ai\oradata\FREE\pdb_example2\');
```

After creating it, we make sure it's open, visualize the path for their datafiles and finally delete it. From now on, we will work with the pdb_example1.

3.User creation, privileges, roles and user profiles

The following task is to create some users, grant some privileges and roles and assign profiles to improve security and resource management.

First, we will create a common user, to create this user we have to be in the container database. We will use the sys user for this task.

```sql
--Check out the parameter common_user_prefix
show parameter common_user_prefix;--VALUE=C##

--Creating a common user
CREATE USER C##supervisor_user
IDENTIFIED BY oracle;

--Check if the user was created
SELECT username, account_status, profile, common
FROM DBA_USERS
WHERE username='C##SUPERVISOR_USER';--username must be in capital letters!
```

To carry out the following tasks we have to move to the pluggable database pdb_example1, still using the user sys. We will create some users, roles and user profiles. All these things can be seen in the sql file called project1.sql.

4.Changing database parameters

First, we will use the dynamic view V$PARAMETER to check out the modifiable-level for each parameter. We will choose two parameters to modify:

```sql
--Visualize how parameters are modifiable
SELECT name,value,isses_modifiable , issys_modifiable , ispdb_modifiable
FROM V$PARAMETER
ORDER BY name;
```

We are changing the parameters nls_date_format in session level and max_idle_time in system level.

```sql
--nls_date_format value=null->derived parameter ISSES_MODIFIABLE=TRUE IS
ALTER SESSION set nls_date_format= 'dd-month-yyyy';
--The value for this session has changed
SELECT SYSDATE FROM dual;
SELECT name,value,isses_modifiable , issys_modifiable , ispdb_modifiable
FROM V$PARAMETER
WHERE name='nls_date_format';

--max_idle_time
SELECT *
FROM V$PARAMETER
where name='max_idle_time';

--These changes will be apply in memory and in the spfile
ALTER SYSTEM set max_idle_time=20 scope=both;

--Reset parameter file
ALTER SYSTEM RESET max_idle_time scope=both;
--The value we set up before in the spfile disappeared
```

We can check the documentation to find out about the parameters we are interested in:

NLS_DATE_FORMAT

Property	Description
Parameter type	String
Syntax	`NLS_DATE_FORMAT = "format"`
Default value	Derived from `NLS_TERRITORY`
Modifiable	`ALTER SESSION`
Range of values	Any valid date format mask but not exceeding a fixed length
Basic	No

2.224 MAX_IDLE_TIME

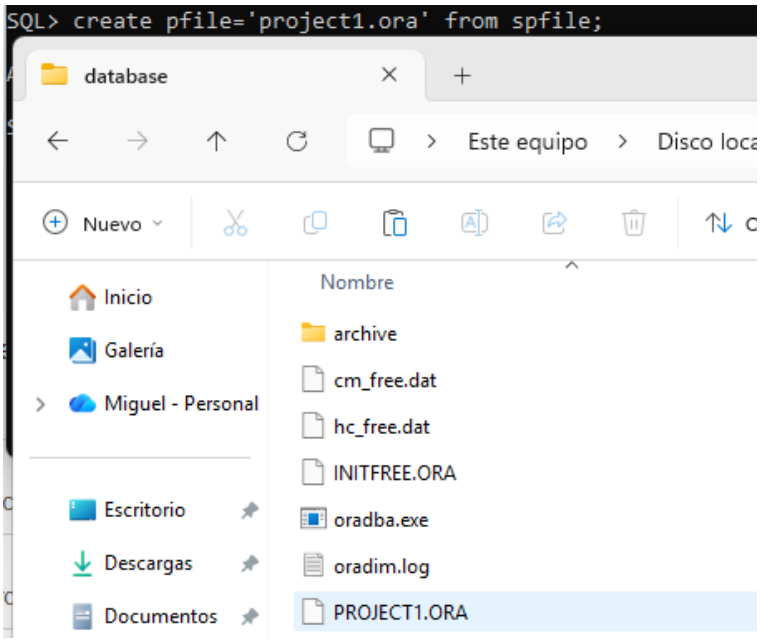
`MAX_IDLE_TIME` specifies the maximum number of minutes that a session can be idle. After that point, the session is automatically terminated.

Property	Description
Parameter type	Integer
Default value	0
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	0 to the maximum integer. The value of 0 indicates that there is no limit.
Basic	No

If we check the spfile, we will find a value for the parameter max_idle_time that it wasn't there before:

```
*.dispatchers='(PROTOCOL=TCP)(DI
*.enable_pluggable_database=true
*.local_listen□C  C"  □
*.max_idle_time=20
*.max_shared_servers=20
*.nls_language='SPANISH'
*.nls_territory='SPAIN'
*.open_cursors=410
```

To conclude with this part we are going to create a pfile from the spfile. Unlike the spfile, the pfile can be modified manually. We will set up a new value for the max_idle_time.



We will shutdown the database and startup it again using the pfile we just created. Also, we can check if we are using a spfile or a pfile checking the value of spfile parameter. If the value is null it means we are using a pfile.

```
SQL> create pfile='project1.ora' from spfile;

Archivo creado.

SQL> show parameter max_idle_time;

NAME                                 TYPE        VALUE
------------------------------------ ----------- ------------------------------
max_idle_time                        integer     0
SQL> shutdown immediate;
Base de datos cerrada.
Base de datos desmontada.
Instancia ORACLE cerrada.
SQL> startup pfile=C:\app\sauky\product\23ai\dbhomeFree\database\project1.ora;
Instancia ORACLE iniciada.

Total System Global Area 1603983088 bytes
Fixed Size                  5617392 bytes
Variable Size            1023410176 bytes
Database Buffers          570425344 bytes
Redo Buffers                4530176 bytes
Base de datos montada.
Base de datos abierta.
SQL> show parameter spfile;

NAME                                 TYPE        VALUE
------------------------------------ ----------- ------------------------------
spfile                               string
```

```
SQL> show parameter max_idle_time;

NAME                                 TYPE        VALUE
------------------------------------ ----------- --------
max_idle_time                        integer     15
```

5.Resources

Get info about max_idle_time parameter:
[Max_idle_time]

Get info about nls_date_format parameter:
[nls_date_format]